

Rod Interpretation

What is going on here?

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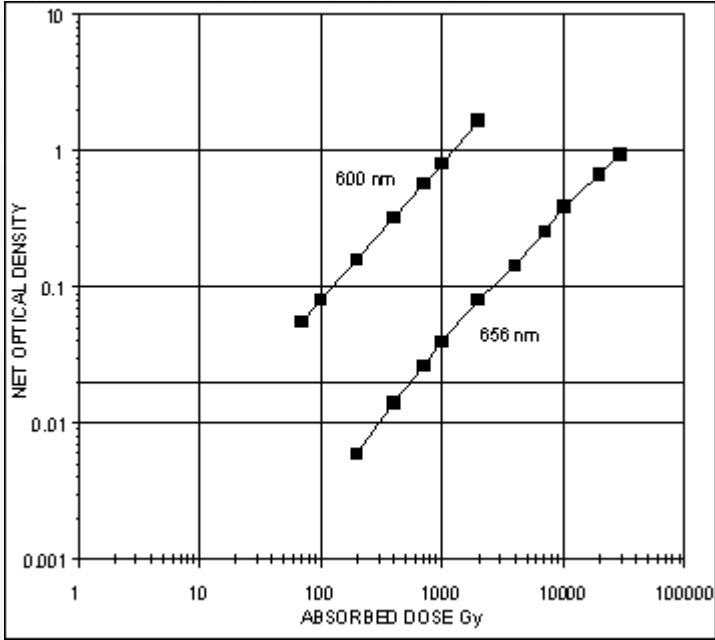
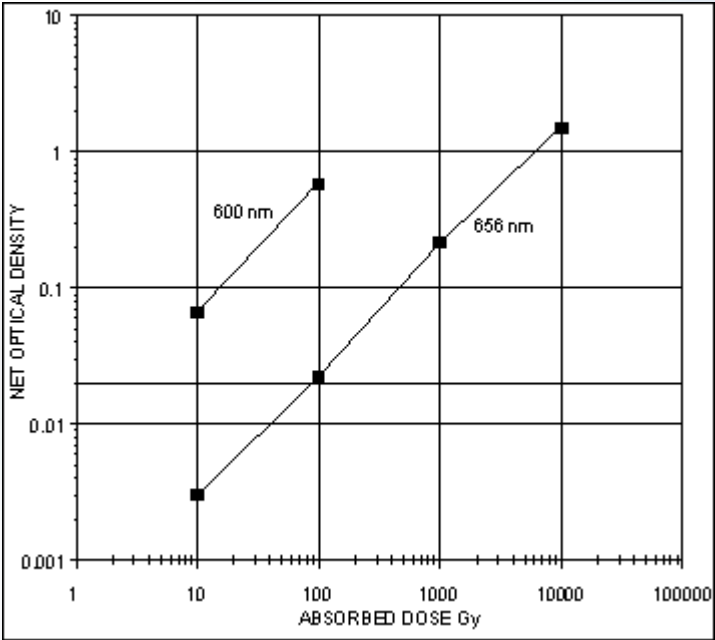


Outline

- Rod Range
 - Theory
 - Complications
- Exemplars
 - Significantly less than 1 krad
 - Less than 1 krad
 - Less than 10 krad
 - Less than 100 krad
 - Greater than 150 krad
- Curiosities
- Summary

Rod Range: Theory

These are the plots from FWT for low dose (top) and high dose (bottom) rods. Based on these plots, you could come up with the following:



	Net Optical Density		Dose (krad)	
	Min	Max	Min	Max
600 nm (low)	0.06	0.6	1	10
656 nm (low)	0.002	1.2	1	1000
600 nm (high)	0.06	2.0	7	200
600 nm (high)	0.006	1.0	20	3000

Rod Range: Complications

- If you have spent some time either reading rods or looking at the data, you will notice that on rods dosed with less than the 1 krad threshold, the net optical density is actually greater than the minimum net optical density suggested by the plots
- An alternative way to look at it – for rods receiving no substantial dose, the dosage calculated is sometimes higher than the lower bound for the rod / wavelength combination
- This suggests that a dose larger than what the plots suggest may be necessary to actually register anything above the noise
- The most reliable way is to look at a few exemplars to help determine applicable range for each of the four cases – low at 600 nm, low at 656 nm, high at 600 nm, high at 656 nm

Exemplar: Significantly less than 1 krad

- This is going to inform the first draft of lower bounds for each rod / wavelength combo
- R-66 (S-5 location), read 03/14/2025

	Optical (600 nm)	Dosage (krad)	Optical (656 nm)	Dosage (krad)
Low Dose Rod 1	0.183	-2.20 (x)	0.71	19.39 (x)
Low Dose Rod 2	0.215	-1.52 (x)	0.72	19.93 (x)
High Dose Rod 1	0.105	7.74 (x)	0.075	110.3 (x)
High Dose Rod 2	0.092	6.20 (x)	0.072	104.9 (x)

- Both TLDs covering this time period did not register a reading above background
- (x) means this readback isn't valid – dose doesn't fall into applicable range
- (?) means this readback validity is uncertain

Draft Boundaries

	Dose (krad)	
	Min	Max
600 nm (low)	1	
656 nm (low)	20	
600 nm (high)	8	
600 nm (high)	110	

- As the other exemplars are presented, the ranges are given in the bottom right
- Changes to the ranges are indicated by arrows, prompted by the readback shown
- Net optical density not shown, since calibration is a routine part of the setup for rod reading

Exemplar: Less than 1 krad

- R-55 (N19 location, misread as N17), read 03/14/2025

	600 nm Dosage (krad)	656 nm Dosage (krad)
Low Dose Rod 1	0.95 (?)	21 (x)
Low Dose Rod 2	0.48 (?)	23 (x)
High Dose Rod 1	9.0 (x)	90.5 (x)
High Dose Rod 2	11.29 (x)	122.8 (x)

- One of the TLDs covering this time period saturated for photons, ie, dose > 1 krad
- Maybe indicates that 1 krad lower bound should be instead 2 krad

	Dose (krad)	
	Min	Max
600 nm (low)	1	
656 nm (low)	20 → 24	
600 nm (high)	8 → 12	
600 nm (high)	110 → 123	

Exemplar: Less than 10 krad

- R-8 (Y16 location), read 03/14/2025

	600 nm Dosage (krad)	656 nm Dosage (krad)
Low Dose Rod 1	2.04	26.9 (x)
Low Dose Rod 2	2.51	29.6 (x)
High Dose Rod 1	10.34 (x)	108.5 (x)
High Dose Rod 2	7.74 (x)	92.3 (x)

- One of the TLDs covering this time period saturated for photons, ie, dose > 1 krad
- In comparison, that TLD had a higher neutron dose than the saturated TLD for R-55, but I’m pretty sure we can’t assume that the neutron / gamma doses are correlated, so that’s not really helpful...

	Dose (krad)	
	Min	Max
600 nm (low)	1 → 2	
656 nm (low)	24 → 30	
600 nm (high)	12	
600 nm (high)	123	

Exemplar: Less than 100 krad

- R-85 (N15 location), read 03/19/2025

	600 nm Dosage (krad)	656 nm Dosage (krad)
Low Dose Rod 1	40.6 (x)	66.0
Low Dose Rod 2	39.7 (x)	59.6
High Dose Rod 1	72.0	131.8 (x)
High Dose Rod 2	68.9	131.8 (x)

- Both TLDs covering this time period were saturated for photons, so at minimum, dose > 2 krad

	Dose (krad)	
	Min	Max
600 nm (low)	2	39
656 nm (low)	30	
600 nm (high)	12	
600 nm (high)	123 → 132	

Exemplar: Greater than 100 krad

- R-135, read 04/10/2025

	600 nm Dosage (krad)	656 nm Dosage (krad)
Low Dose Rod 1	41.9 (x)	184.5
Low Dose Rod 2	42.0 (x)	172.2
High Dose Rod 1	166.0	190.9
High Dose Rod 2	166.5	194.5

	Dose (krad)	
	Min	Max
600 nm (low)	2	39
656 nm (low)	30	
600 nm (high)	12	
600 nm (high)	132	

Draft Boundaries

	Dose (krad)	
	Min	Max
600 nm (low)	2	39
656 nm (low)	30	
600 nm (high)	12	
600 nm (high)	132	

Curiosities: 20-40 krad

- R-138, read 04/16/2025

	600 nm Dosage (krad)		656 nm Dosage (krad)
Low Dose Rod 1	22.25	(?)	40.83
Low Dose Rod 2	20.93	(?)	37.08
High Dose Rod 1	33.16		101.29 (x)
High Dose Rod 2	34.82		124.60 (x)

- Is the ~20 krad reading valid or beyond the bounds of the low dose 600 nm range?
- If it's beyond the range, the max should be set ~20

	Dose (krad)	
	Min	Max
600 nm (low)	2	39
656 nm (low)	30	
600 nm (high)	12	
600 nm (high)	132	

Curiosities: Strapped to a beam loss monitor

- ILX9S00, installed 04/23/2025, pulled 05/07/2025, read 05/09/2025

	600 nm Dosage (krad)	656 nm Dosage (krad)
Low Dose Rod 1	40.77 (x)	75.13
Low Dose Rod 2	41.15 (x)	76.74
High Dose Rod 1	75.85	126.39 (x)
High Dose Rod 2	76.08	124.60 (x)

- Strapped to IIX9S00
- Integrating on the readback gets 51 krad
 - on one hand, the right order of magnitude
 - on the other, there's not much variation in the rod readback, so I'm a bit skeptical
 - as a caveat, the monitor saturated, so the value is closer to > 51 krad, with no upper limit

	Dose (krad)	
	Min	Max
600 nm (low)	2	39
656 nm (low)	30	
600 nm (high)	12	
600 nm (high)	132	

Curiosities: Strapped to a beam loss monitor, ~20 krad

- ILX7S01, pulled 05/07/2025, read 05/09/2025

	600 nm Dosage (krad)		656 nm Dosage (krad)	
Low Dose Rod 1	14.51	(?)	33.33	(x)
Low Dose Rod 2	14.89	(?)	29.04	(x)
High Dose Rod 1	22.40		95.91	(x)
High Dose Rod 2	22.64		104.88	(x)

- Strapped to IIX7S01
- Integrating on the readback gets 6.92 krad
-- questions about these ion chambers increase
- Is the ~15 krad reading valid or beyond the bounds of the low dose 600 nm range?
- If it's beyond the range, the max should be set ~14

	Dose (krad)	
	Min	Max
600 nm (low)	2	39
656 nm (low)	30	
600 nm (high)	12	
600 nm (high)	132	

Summary

- While the manufacturer provided ranges don't match up with recent data acquisition, the amount of data has provided at least a rough idea of applicable ranges for each dose/wavelength combination
 - With the exception of the maximum limit on the 600 nm low dose reading – is it 10, 20, 30 krad?
 - The upper limits of the others aren't relevant for our studies, but we could probably explore the 200 krad limit of the high dose 600 nm if it would be helpful
- Not pleased by the 2 krad minimum of the 600 nm low dose, but the TLD saturation provides a better error bar than some of the higher dose rods, so it could be worse
- Not sure if I'm being overly critical of the discrepancies between the dosimetry and the ILX monitors (ILX3S00, ILX7S01, ILX9S00) – thoughts? Will start comparing NDX data next

QUESTIONS?